

Taiwan Explorers

GROUP 18 FLOWERS

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GITHUB REPOSITORY LINK

<https://github.com/CMPT-276-SUMMER-2025/final-project-18-flowers.git>

Project Proposal

1. Project Overview

- **What is the project?**

A web app designed to make exploring and planning a trip to Taiwan simple and accessible for users all across the world.

- **What problems does the project aim to solve?**

As of 2024, Taiwan was ranked 5th in Asia for international tourism and is expecting around 10 million tourists in 2025, which is a 27.4% increase from the previous year. However, despite its growing popularity, it still lacks dedicated travel guides compared to destinations like Japan or South Korea. This project aims to fill that gap by building a website that helps people plan their trips and connect with everything Taiwan has to offer.

Sources:

- https://focustaiwan.tw/business/202504020024?utm_source=chatgpt.com
- <https://www.ocac.gov.tw/OCAC/Eng/Pages/Detail.aspx?nodeid=329&pid=72535168>

- **Where did the idea come from?**

During our brainstorming session, we agreed that the goal of this project would be to take something that already exists, replicate it, and improve certain aspects with our own twists. Initially, we considered creating a trip planner website for Japan, a popular tourist destination known for its unique culture. However, we quickly realized that the market for Japanese travel sites is oversaturated, with dozens if not hundreds of websites offering similar services. With that in mind, we decided to explore other countries where this idea could stand out. Ultimately, we chose Taiwan because it stood out to us as having surprisingly few dedicated travel websites catered to Western audiences.

- **Why is the project important?**

The importance of this project stems from our goal to showcase the beauty of Taiwan to people who may not have heard of it as much compared to countries like Japan and

South Korea. We aim to offer a service through our web application that shares useful information, creates well-structured travel plans, and makes the process of planning a trip easy and smooth.

- **Who are the potential users (application groups) and what are their needs?**

The potential users of our website are potential travelers interested in visiting Taiwan. These users need reliable and easy to find information to help them plan their trips efficiently. This includes details about accommodations, food spots, shopping areas, tourist attractions, estimated costs, and transportation. Ultimately, their main need is to access accurate and the latest travel information that make it easier to create fun, informed, well-structured travel plans tailored to their personal interests and budget.

- **How does the app fulfill the needs?**

To fulfill the needs of potential users, we will implement various features that are essential for travel planning. First and foremost, our website will provide the latest information about Taiwan, including important news, weather updates, earthquake warnings, and more. Secondly, we will add a dedicated section for destinations across Taiwan, offering all the key details travelers need to know. In addition, we will include a section that highlights various interests and activities available in Taiwan to help users discover what suits them best. Lastly, we plan to implement a feature that lets users build personalized travel plans directly through our web app, including destinations, estimated costs, transportation options, routes, accommodations, and more.

2. Personas

Persona 1 – Student / Business Reasons

- **Needs:** A quick and budget-friendly trip with minimal focus on tourist attractions
- **Tech Experience:** High
- **Pain Point:** Limited time and money for trip planning and activities

Persona 2 – Elderly Traveler

- **Needs:** A simple, slow-paced itinerary with minimal physical activity
- **Tech Experience:** Low to Medium
- **Pain Point:** Dislikes overly complex plans and long walking distances

Persona 3 – Tourist Explorer

- **Needs:** Access to major tourist attractions, a variety of restaurants, and flexible spending options
- **Tech Experience:** Medium
- **Pain Point:** Needs detailed, precise, and well-organized trip planning tools / features

Persona 4 – Hiker / Nature Enthusiast

- **Needs:** Outdoor trails, scenic spots, and walking routes
- **Tech Experience:** Varies
- **Pain Point:** Travel plans are highly dependent on weather conditions and route planning

3. List of Chosen APIs, Features, and User Stories

API 1 (**Primary**): Google Maps API

API Link: [Google Maps Platform Documentation](#) | [Google for Developers](#)

Features

- **Feature 1:** Build dynamic and interactive maps with integrated features such as map events, controls, panning and zooming, map types, and customizability.
- **Feature 2:** Show static maps of various locations that include markers, custom styling, different map types, and customizable paths between points.
- **Feature 3:** Incorporate 360 degree street view to maps.

User Stories

- As a first time traveler to Taiwan, I want to be able to view an interactive map so that I can get a feel of where I am in the world.
- As a visual learner, I want to be able to view detailed static maps of the locations that I'm visiting so that I can keep mental images of different points at the destination and feel prepared when I get there.
- As a novice traveler with little experience in Taiwan, I want to be able to have access to street view of the locations that I would like to visit so that I can get a preview of what the place looks like before arriving there.

API 2 (Primary): Google Gemini API

API Link: <https://ai.google.dev/gemini-api/docs>

Features

- **Feature 1:** Smart Trip Planner with Google Services Integration (done by combining Gemini with google maps/places).
- **Feature 2:** Chatbot/FAQ Assistant used to any answer user questions with the help of AI assistant
- **Feature 3:** Custom Budget Trip Ideas, which generates a budget based on the budget, city and days entered.

User Stories

- As a busy business man travelling to Taiwan, I want a good trip planner when I arrive in Taiwan so that I don't need to waste time on trying to find routes or locations while in Taiwan.
- As a first time tourist to Taiwan, I want all my possible questions about Taiwan answered so that there is no room for any confusion when I arrive there.
- As a student with a tight budget, I want to be able to plan trips to Taiwan so that it doesn't exceed my budget limit.

API 3 (Backup): Google Places API

API Link:

<https://developers.google.com/maps/documentation/places/web-service/overview>

Features

- **Feature 1:** Display location and information data about various attractions, accommodations, and more.
- **Feature 2:** Add a place search feature where users can input text to find nearby destinations data based on various factors like food, name, and area.
- **Feature 3:** A feature that enables users to retrieve photos, costs if applicable, and reviews for each place, helping them better decide where to visit based on real visitor feedback and visuals.

User Stories

- As a frequent tourist, I want to be able to plan out my trips with the best a country has to offer so that I can create memorable memories and have a blast.

- As an enthusiastic first time visitor, I want to explore specific attractions and places near destinations to learn more about the places I plan to go and can visit.
- As a cautious planner, I want to see reviews and photos of places so that I can make informed decisions before visiting to maximize the day.

API 4 (Backup): Google Directions API

API Link: <https://developers.google.com/maps/documentation/directions>

Features

- **Feature 1:** The Directions API calculates routes between a series of way-points and returns directions for the chosen transportation such as transit, driving, walking, or bicycling.
- **Feature 2:** User provides parameters which the API uses to avoid undesirable routes including tolls, tracks, ships.
- **Feature 3:** Given departure time, it calculates travel distance and traffic to predict arrival time.

User Stories

- As an athletic traveler, I want to choose between a variety of transportation so that I can enjoy hiking, biking and mountain-climbing on my trip.
- As a person who suffers from motion-sickness, I want to inform the route planner so that it avoids bumpy roads in my route.
- As an employee with a busy schedule, I want the application to take heavy traffic and construction into account so that I prepare a sufficient amount of traveling time.

4. Low-fidelity Storyboard (Screenshots + Link)

Figma Link:

<https://www.figma.com/design/sn5cppElkfkMjcGlOqoYrF/18-Flowers?m=auto&t=8oWMgOitxfaW52bX-1>

Screenshots of Storyboard:

Home page screenshots:

Taiwan Explorers

HomeDestinationsInterestsPlan a trip

Search

Photo of Taiwan

Explore
Taiwan
Plan your trip now!

Destinations

Image of an Destination

Image of an Destination

Image of an Destination

Image of an Destination

See more destinations

plane

train

hotels

...

Button that leads to external website
Text on button: "Submit"

Redirects to external websites. Ex. bookings.com

Interests

iconWhere to go sightseeing

iconWhat to eat

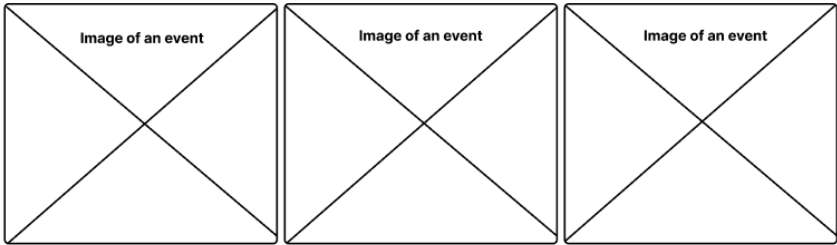
iconWhere to go shopping

iconWhere to book hotel

See more interests

Image of an attraction

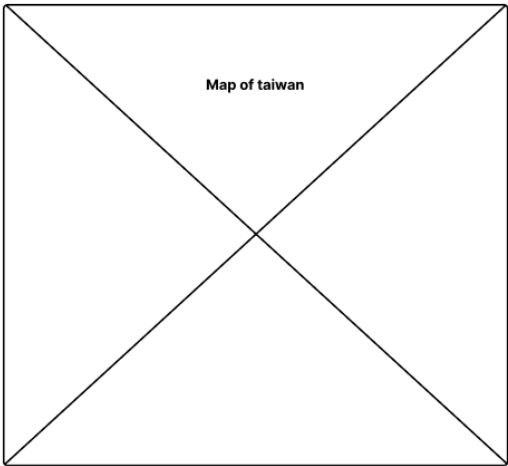
Top Attractions



Story 1

Story 2

Story 3



Plan Your Trip

Review 1

Name 1
Description

Review 2

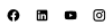
Name 2
Description

Review 3

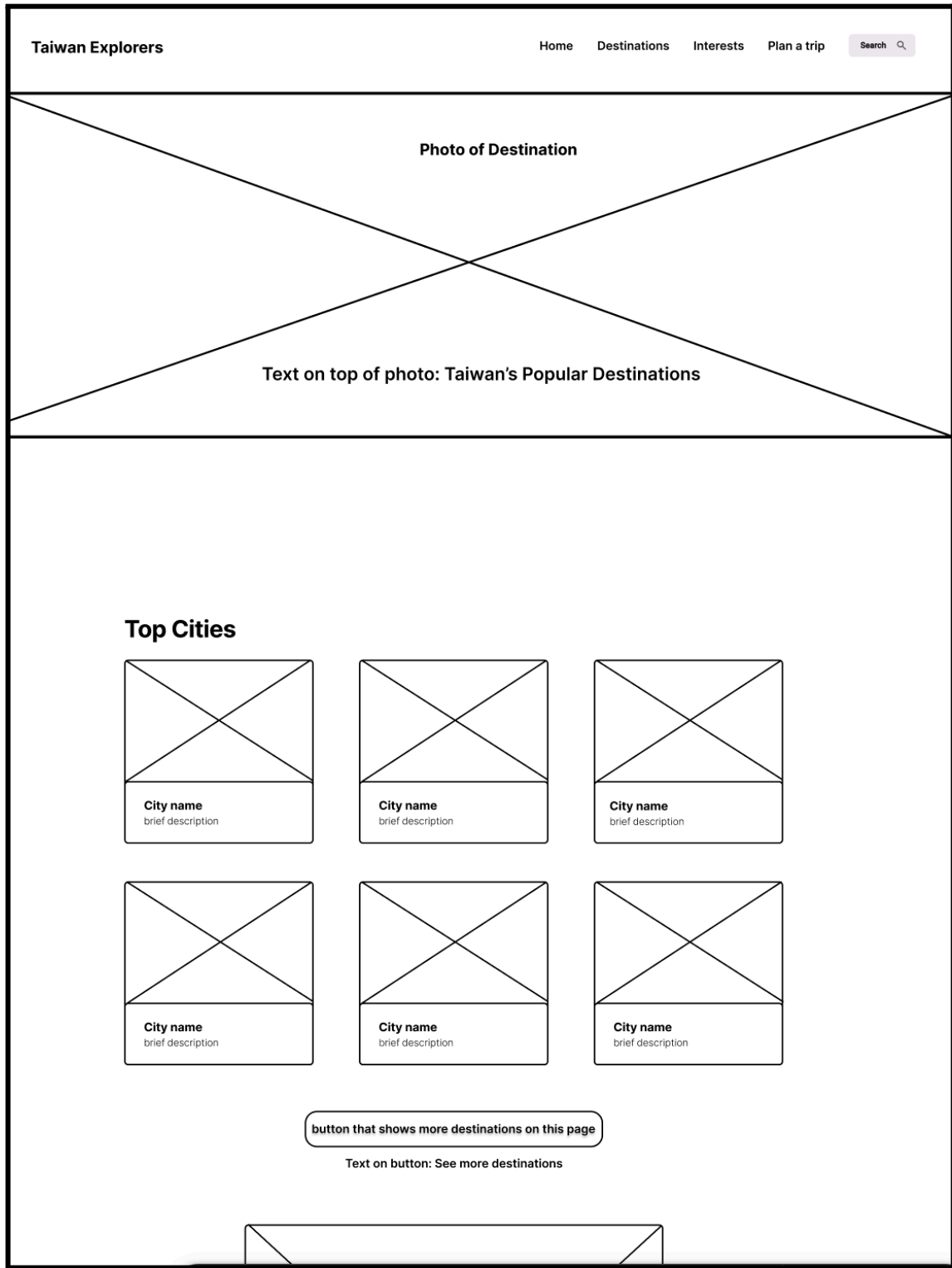
Name 3
Description

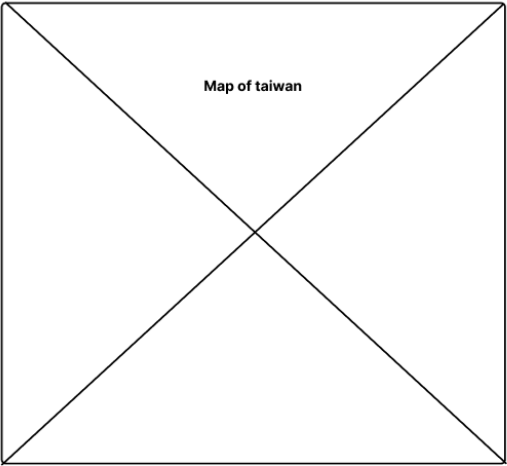
Top 10 Essentials

Description



Destinations screenshots:





Taiwan Explorers



Topic

Page

Page

Page

Topic

Page

Page

Page

Topic

Page

Page

Page

Interests screenshots:

Taiwan Explorers

Home

Destinations

Interests

Plan a trip

Search

Here are things that may meet your interest

Interest Type 1

Image of a landscape

Landscape

Image of a landscape

Landscape

Image of a landscape

Landscape

Image of a landscape

Landscape

Image of a landscape

Landscape

Image of a landscape

Landscape

Interest Type 2

Image of a restaurant Restaurant	Image of a restaurant Restaurant	Image of a restaurant Restaurant
Image of a restaurant Restaurant	Image of a restaurant Restaurant	Image of a restaurant Restaurant

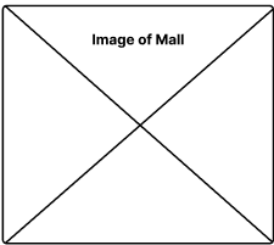
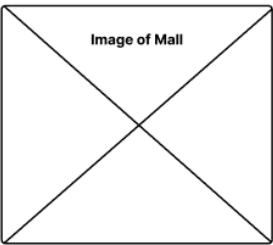
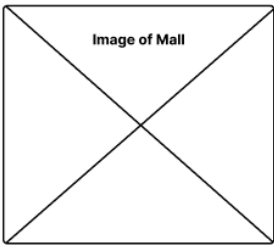
Interest Type 3

Image of Mall	Image of Mall	Image of Mall
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Night Market 1

Night Market 2

Mall

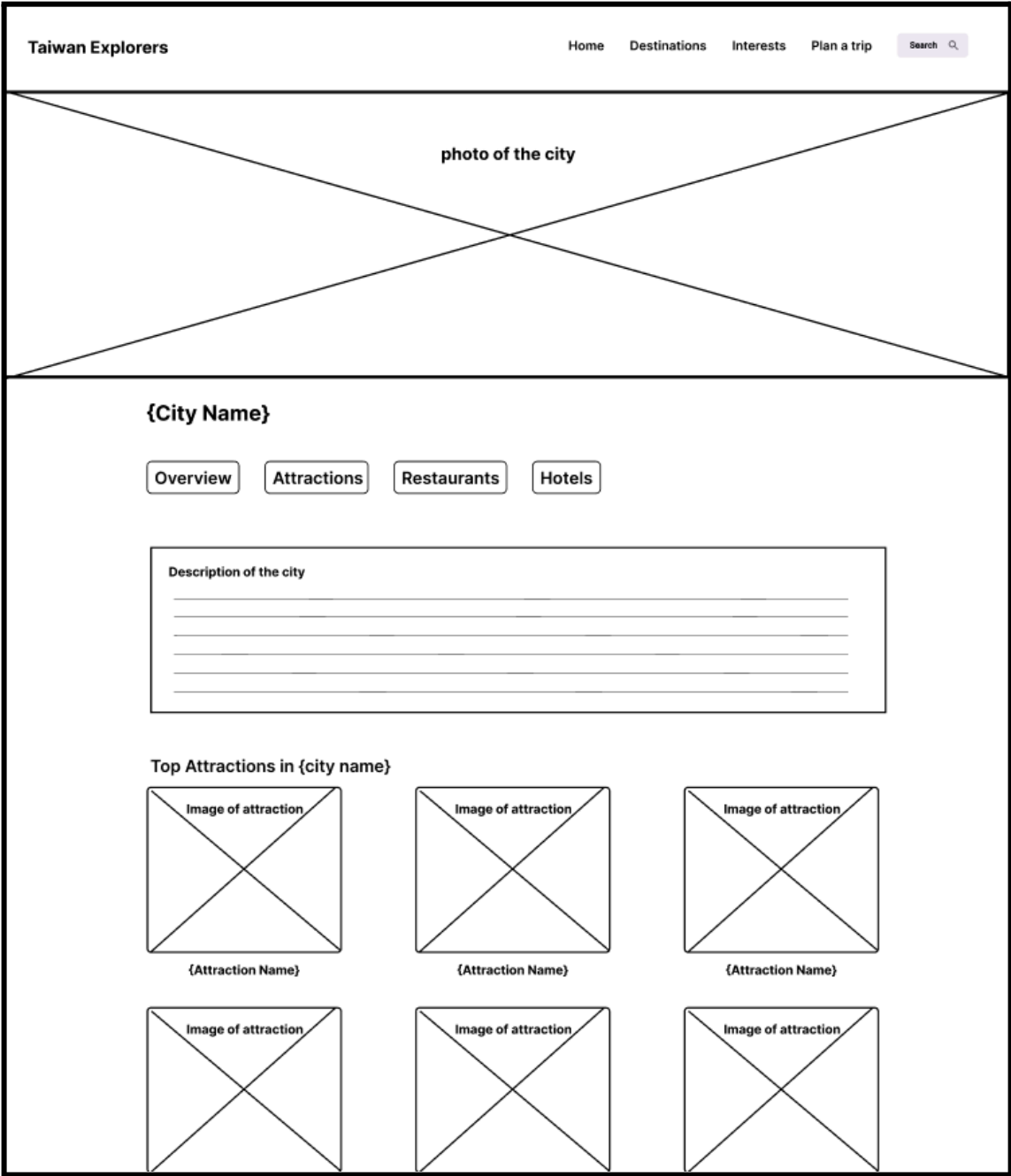


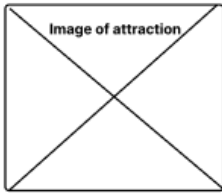
Mall

Mall

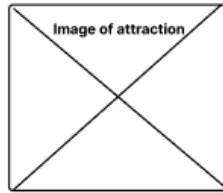
Mall

City Interests screenshots:

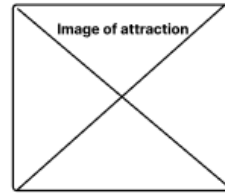




{Attraction Name}



{Attraction Name}



{Attraction Name}

Restaurants around {city name}



Restaurant name

2.2km \$-\$ Restaurant tag



Restaurant name

2.2km \$-\$ Restaurant tag



Restaurant name

2.2km \$-\$ Restaurant tag



Restaurant name

2.2km \$-\$ Restaurant tag

Hotels around {city name}



Hotel name

2.2km \$-\$ Hotel tag



Hotel name

2.2km \$-\$ Hotel tag



Hotel name

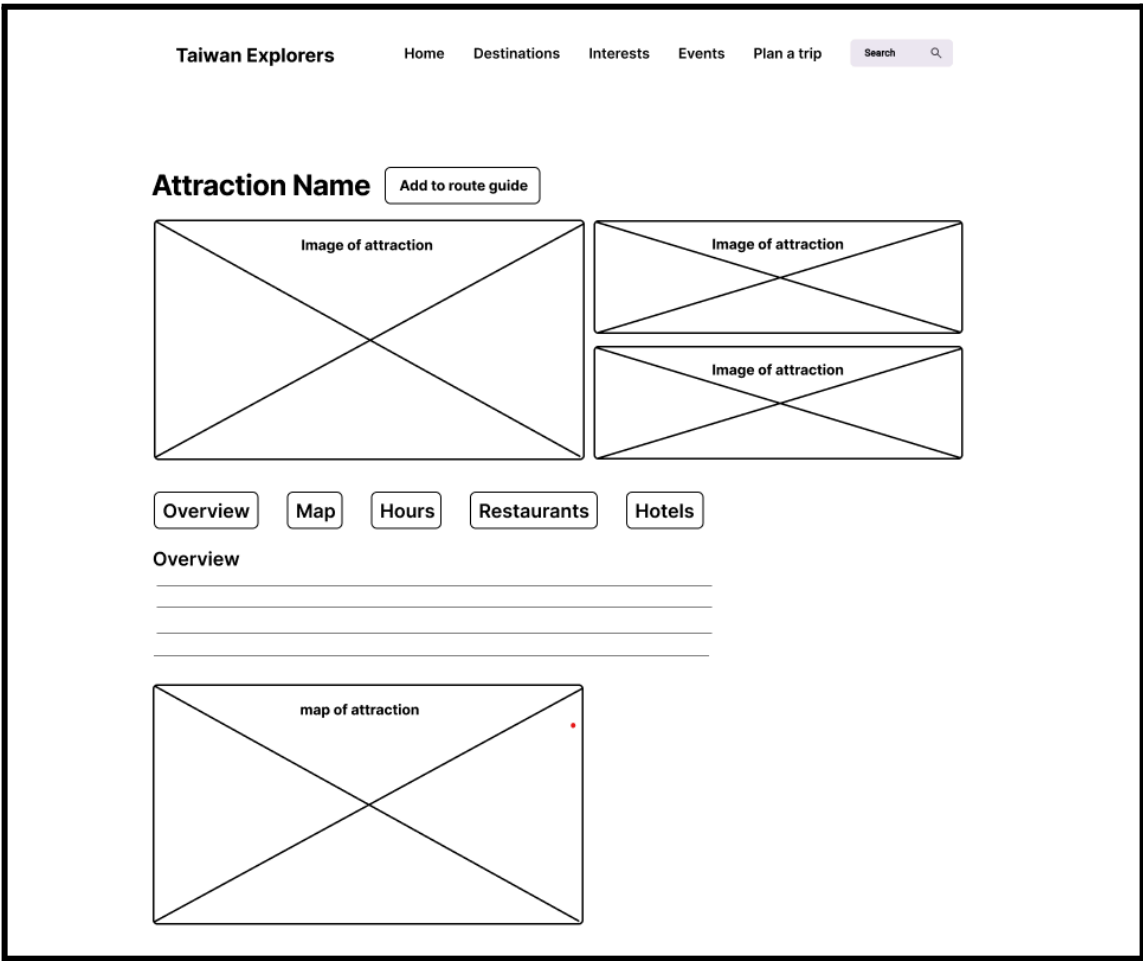
2.2km \$-\$ Hotel tag



Hotel name

2.2km \$-\$ Hotel tag

Attraction info screenshots:



Hours

Monday 8:00-19:00
Tuesday 8:00-19:00
Wednesday 8:00-19:00
Thursday 8:00-19:00
Friday 8:00-19:00
Saturday 8:00-19:00
Sunday 8:00-19:00

Tours

Tour 1 Tour description with link
Tour 2 Tour description with link
Tour 3 Tour description with link
Tour 4 Tour description with link

Closest Restaurants

	Restaurant name 2.2km \$-\$\$ Restaurant tag
	Restaurant name 2.2km \$-\$\$ Restaurant tag
	Restaurant name 2.2km \$-\$\$ Restaurant tag
	Restaurant name 2.2km \$-\$\$ Restaurant tag
	Restaurant name 2.2km \$-\$\$ Restaurant tag

Closest Hotels

	Hotel name 2.2km \$-\$\$ Restaurant tag
	Hotel name 2.2km \$-\$\$ Restaurant tag
	Hotel name 2.2km \$-\$\$ Restaurant tag
	Hotel name 2.2km \$-\$\$ Restaurant tag
	Hotel name 2.2km \$-\$\$ Restaurant tag

Travel Guide screenshots:

Taiwan Explorers

Home

Destinations

Interests

Plan a trip

Search

Plan your own personal travel plan here

Where are you going?

Country

↓

How many days?

of days

↓

Interests

Food/Culture/etc.

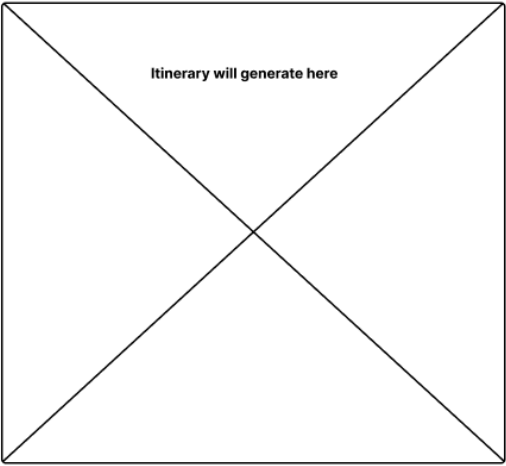
↓

Add specific times?

Yes / No

↓

Generate Itinerary

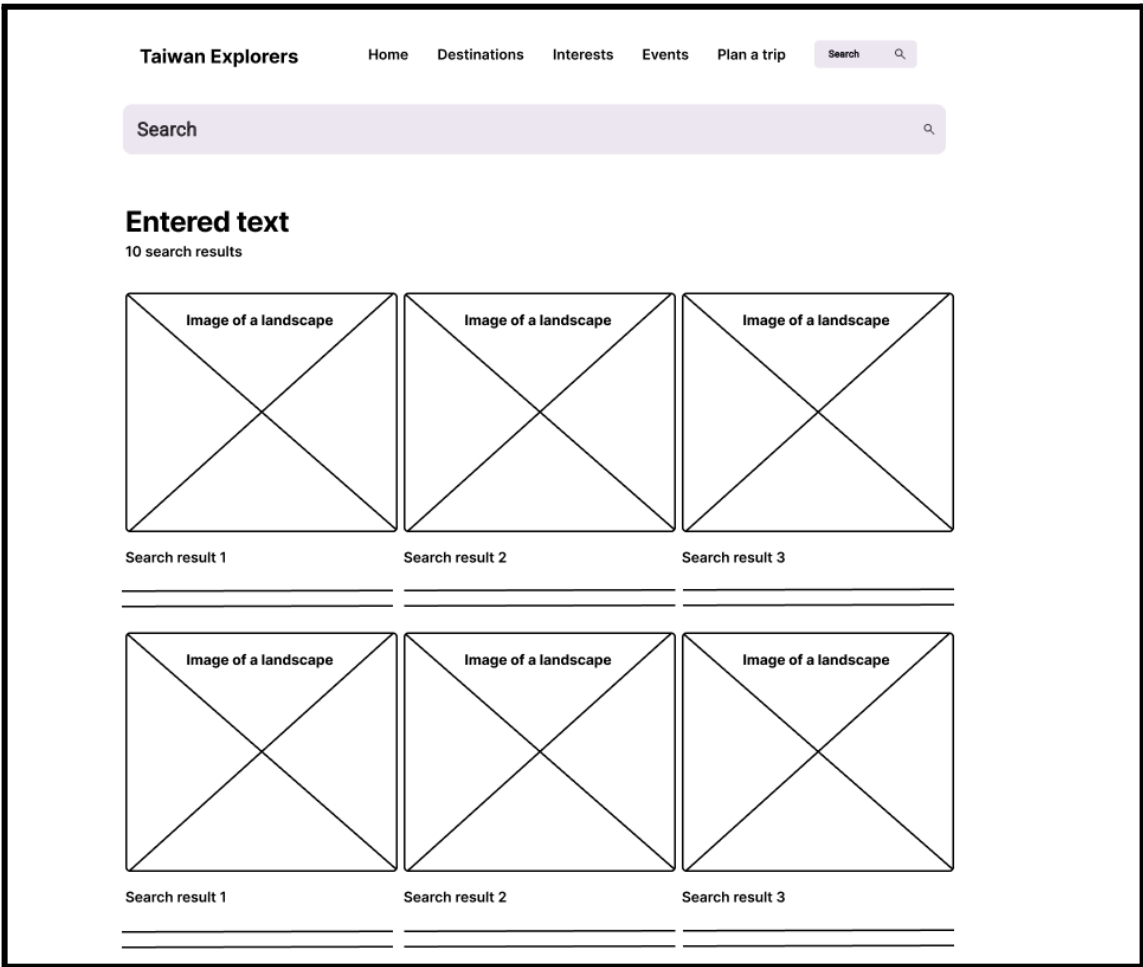


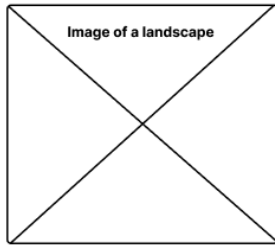
Taiwan Explorers



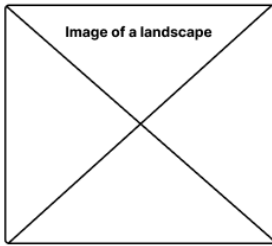
Topic	Topic	Topic
Page	Page	Page
Page	Page	Page
Page	Page	Page

Search page screenshots:

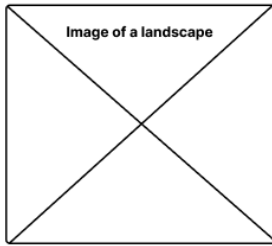




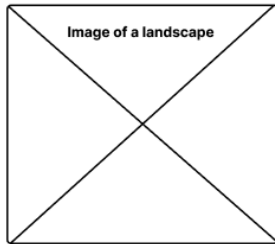
Search result 1



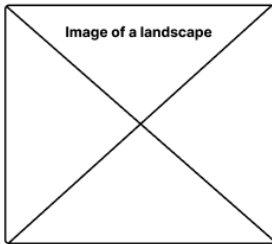
Search result 2



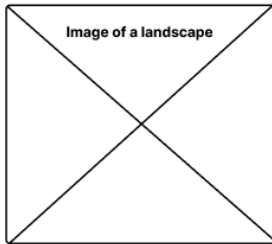
Search result 3



Search result 1



Search result 2



Search result 3

5. Technology Stack

HTML

While HTML is the foundation of web development, it will play a minimal role in our project since we're using React with JSX, which allows us to structure UI components using a syntax that closely resembles HTML. The main use of pure HTML will be for setting up essentials such as the website's **favicon** and **title** in the `<head></head>` section.

CSS

CSS will support Tailwind CSS by handling reusable styles that benefit from defined classes/IDs such as buttons, box components, and theme variables (e.g., colors & fonts reflecting Taiwanese culture or branding). Additionally, CSS's grid and flexbox layouts will help in building responsive and visually structured layouts across all screen sizes.

Tailwind CSS

Tailwind will be our main styling utility due to its efficiency and responsiveness. We'll use it to rapidly prototype and style unique components like the navigation menu, and to reduce CSS file clutter. Its responsive breakpoints and utility classes will help make the website friendly across devices of varying screen sizes.

JavaScript

As the backbone of React, JavaScript will be central to adding interactivity and integrating dynamic content. We'll use it to fetch and render data from external APIs (e.g., tourist spots, weather, or cultural events in Taiwan), manage user interactions, and implement UI effects like modals or animations.

React

React was chosen for its component-based architecture, which aligns well with the modular nature of our Taiwan-themed site. Key features like the attractions display, navigation bar, and interactive maps will be built as reusable components. Additionally, React's ecosystem, including animation libraries like AOS, allows us to add polished touches and enhance the overall user experience.